

EC151

Basics of Electronics Engineering Lab

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Section: B1

Experiment Number	Aim
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Basics of Electronics Engineering LabExperiment - 01

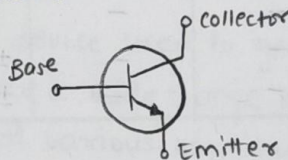
Aim: To identify active and passive elements in a circuit and to familiarize with basic electronic instruments.

Apparatus: Resistors, Capacitors, Transformers, Diodes, Transistors.

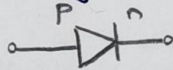
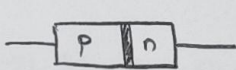
Theory:

→ Active Elements: These are elements capable of delivering energy to the circuit. They require an external power source to operate and control current flow.

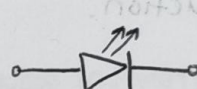
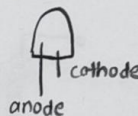
i) Transistor: It is a semiconductor device used to amplify or switch electrical signals and power.



ii) Diodes: It is a 2-terminal semiconductor device that allows current flow in a single direction.



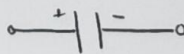
P-n junction diode



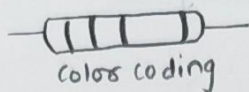
LED

→ Passive Elements: These are elements not capable of delivering energy to the circuit. They do not require an external power source and cannot amplify energy. They can only store, dissipate and maintain energy in a circuit.

i) Capacitor: It is a device that stores energy in the form of an electric field between two conductive plates separated by a finite distance with a dielectric.

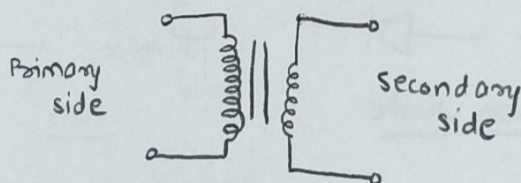


ii) Resistor: It is a circuit element that opposes current flow.



Color	Digit	Multiplier	Tolerance
Black	0	$\times 10^0$	—
Brown	1	$\times 10^1$	—
Red	2	$\times 10^2$	—
Orange	3	$\times 10^3$	—
Yellow	4	$\times 10^4$	—
Green	5	$\times 10^5$	—
Blue	6	$\times 10^6$	—
Violet	7	$\times 10^7$	—
Gray	8	$\times 10^8$	—
White	9	$\times 10^9$	—
Gold	—	—	$\pm 5\%$
Silver	—	—	$\pm 10\%$
No color	—	—	$\pm 20\%$

iii) Transformer: It is an electromagnetic device that transfers electrical energy between two or more circuits via electromagnetic induction.



Conclusion: Active and passive elements are studied and familiarized.

Date: 16/01/25

Experiment - 02

Aim: To study various electrical equipments - multimeter, power supply, function generator, DSO.

Apparatus: Multimeter, power supply, Function Generator, DSO.

Theory:

- Multimeter: It is a versatile electronic device used to measure various electrical properties, like voltage, current, resistance, etc. It is also used to diagnose and test electrical circuits, both AC and DC.
- Power supply: It is an electrical device that supplies electric power to an electrical load. It converts the power from the source to the correct format (voltage / current) and magnitude. It is also used to test devices by varying voltages for performance.
- Function Generator: It is a device used to generate different types of electrical waveforms over a wide range of frequency. It can also generate waveforms of various amplitudes, and other parameters according to our need. Some of the waveform include sine, square, pulse, random, and noise.
- Digital Storage Oscilloscope (DSO): It is an electrical instrument that measures and records electrical signal. It converts analog signal into digital form and store it for easy recall and analysis. It displays waveforms and provides multiple channels to examine signals. The output of a function generator can be used as a DSO input.

Conclusion: Various electrical equipments like multimeter, power supply, function generator and DSO are studied.

Aim: To measure AC/DC voltage, current, and resistance using a digital multimeter, and various periodic functions using a DSO.

Apparatus: Digital Multimeter, DSO, Resistors, Connecting wires, function generator, etc.

Theory:

- Digital Multimeter (DMM): It is a device used to measure electrical values such as voltage, current, resistance, etc., both AC and DC.
- Digital Storage Oscilloscope (DSO): It is a device used to capture, display, and analyse analog waveforms (of electronic signals) in a digital format. It also stores the data for easy recall and analysis. The output of a function generator can be a DSO input.

Procedure:

→ Measurement of voltage:

- 1) To measure DC voltage, turn the DMM dial to $\overline{\overline{V}}$.
- 2) To measure AC voltage, turn the DMM dial to $\tilde{V}$.
- 3) Connect the black probe to COM and the red one to (V Ω mA).
- 4) Select the required range (V, mV, etc.).
- 5) Measure the voltage by connecting the black probe to (-ve) and the red probe to (+ve) terminals respectively.

→ Measurement of current:

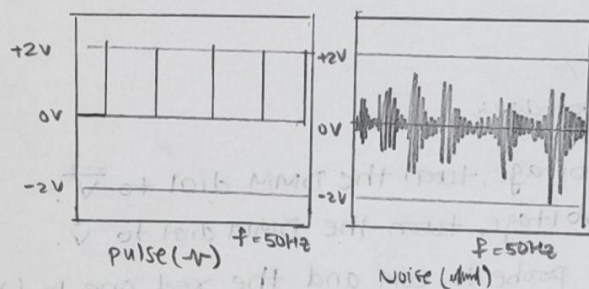
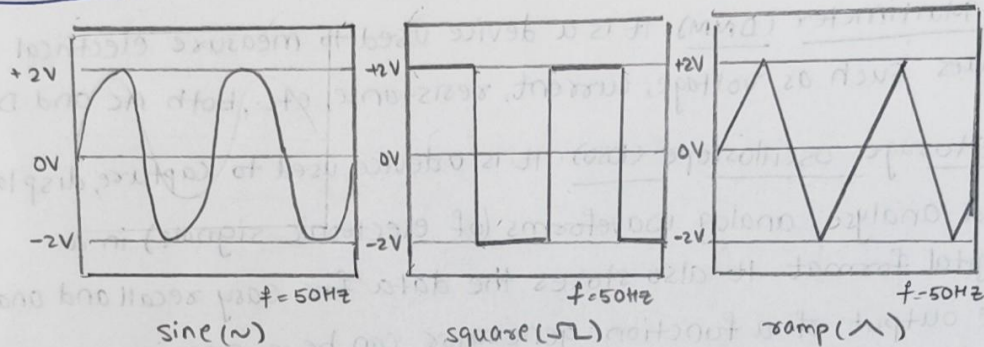
- 1) To measure DC current, turn the DMM dial to $\overline{\overline{A}}$.
- 2) To measure AC current, turn the DMM dial to $\tilde{A}$.
- 3) Connect the probes in a similar way to measure voltage.
- 4) Connect the probes in series to measure current.

→ Getting waveform onto a DSO:

- 1) Connect the DSO to a function generator using connecting wires.
- 2) Turn on the power supply to both DSO and function generator, and turn on both the devices.

- 3) Select the sine wave ($\sim$) from the function generator.
- 4) Set the input frequency of 50Hz (or any desired value) and amplitude of 2Vpp (or any desired value).
- 5) Press the 'Auto Set' button on the DSO and observe the waveform.
- 6) Using the cursors, check the time period and amplitude on DSO.
- 7) Repeat the same for other waveforms.

Observations:



- Conclusion :
- (a) AC / DC Voltages, currents, and resistances have been measured using a Digital Multimeter (DMM).
 - (b) Various waveforms (sine, square, ramp, pulse, noise) have been observed using a DSO.

Aim: To study and verify truth tables for different logic gates.

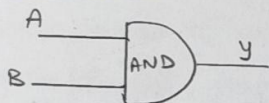
Apparatus: AND gate (SN74F08N), OR gate (74HCT32N), NOR gate (74HC02N), NAND gate (MC74AC00N), NOT gate (SN74HC04N), XOR gate (SN74CH86N), XNOR gate (74HC266N), digital IC kit, connecting wires.

Theory:

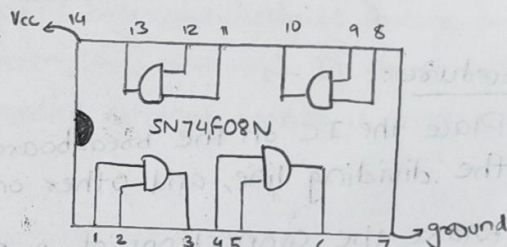
→ Logic Gate: A logic gate is a device that performs boolean function, a logical operation performed by one or more binary inputs that produces a single binary output.

→ Uses of Logic Gates: Logic gates are used in various electronic devices, including calculators to perform tasks like arithmetic operations, data processing, and decision making.

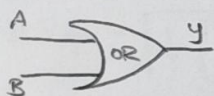
(i) AND Gate:



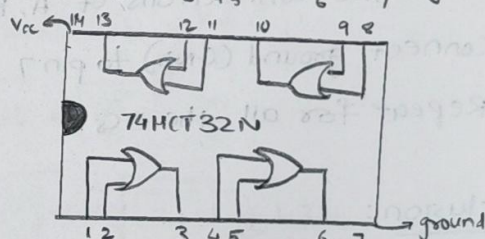
A	B	$Y = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1



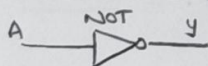
(ii) OR Gate:



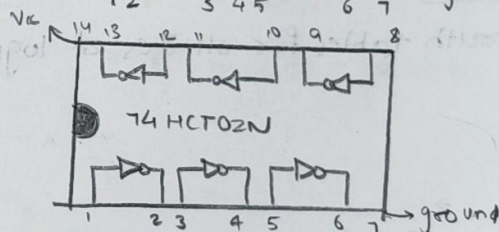
A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1



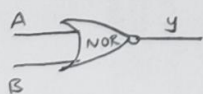
(iii) NOT Gate:



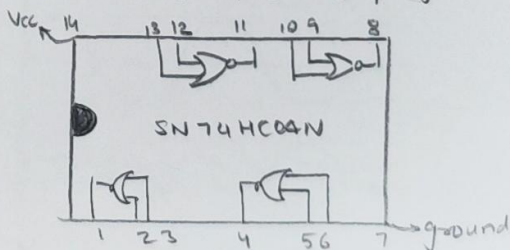
A	$Y = \bar{A}$
0	1
1	0



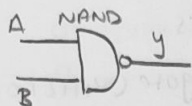
(iv) NOR Gate:



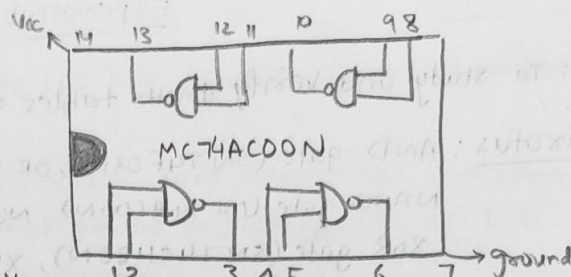
A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0



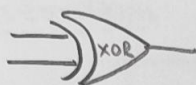
(v) NAND Gate:



A	B	$Y = \overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

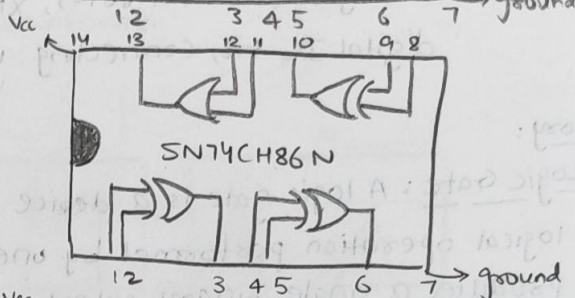


(vi) XOR Gate:

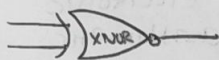


$$Y = A\overline{B} + B\overline{A}$$

A	B	$Y = A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

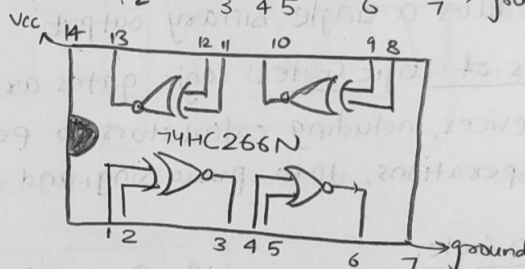


(vii) XNOR Gate:



$$Y = AB + \overline{A}\overline{B}$$

A	B	$Y = A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1



Procedure:

- 1> Place the IC on the breadboard such that one side is on upper side of the dividing line, and other on the bottom (all legs have different potential)
- 2> Make the connections of A, B, and Y as per above diagrams.
- 3> Connect ground (GND) to pin 7 and V_{cc} (+5V) to pin 14.
- 4> Repeat for all the ICs.

Conclusion:

Truth tables for all ICs of logic gates are verified.

— x —

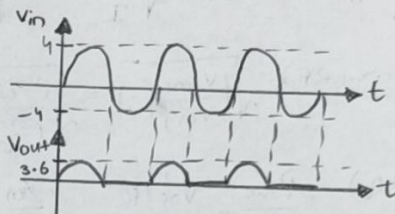
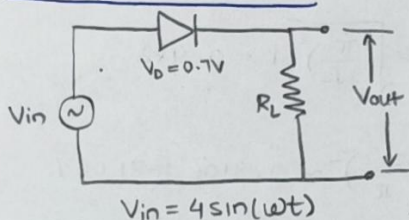
Aim: To study half wave and full wave rectifiers and their characteristics.

Apparatus: step down transformer, diode (IN4007), Resistors, DSO, Function Generator, connecting wires.

Theory:

The process of converting alternating current (AC) to direct current (DC) is called as 'rectification'. The device that rectifies is called a 'rectifier'.

→ Half wave Rectifier:



In the (+ve) half cycle, diode is in forward bias and conducts through the load resistor R_L . Hence, voltage out is seen ($V_{out} \approx V_{in} - V_D$). In the (-ve) half cycle, diode is in reverse bias and doesn't conduct. Hence, it acts as an open circuit, and $V_{out} = 0$.

(a) Peak Inverse Voltage (PIV) = V_m

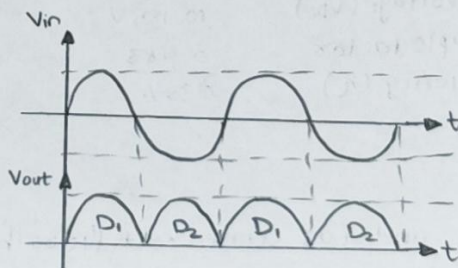
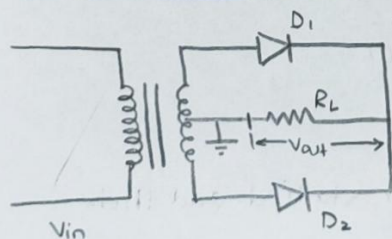
$$(b) V_{avg} = V_{DC} = \frac{1}{T} \int_0^T V_{in}(t) dt = \frac{1}{T} \int_0^T V_m \sin(\omega t) dt = \frac{V_m}{\pi}$$

$$(c) V_{rms} = \sqrt{\frac{1}{T} \int_0^T V_{in}^2(t) dt} = \sqrt{\frac{V_m^2}{T} \int_0^T \sin^2(\omega t) dt} = \sqrt{\frac{V_m^2}{4}} = \frac{V_m}{2}$$

$$(d) \text{Ripple Factor (RF)} = \sqrt{\left(\frac{V_{rms}}{V_{avg}}\right)^2 - 1} = \sqrt{\left(\frac{V_m/2}{V_m/\pi}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2}\right)^2 - 1} \approx 1.211$$

$$(e) \text{Efficiency } (\eta) = \frac{P_{out}}{P_{in}} = \frac{V_{dc}^2 / R}{V_{rms}^2 / R} = \frac{(V_m/\pi)^2}{(V_m/2)^2} = \left(\frac{2}{\pi}\right)^2 = 0.4053 = 40.53\%$$

→ Full wave Rectifier:



In the (+ve) half cycle, D_1 is in forward bias and D_2 is in reverse bias, hence current only from D_1 will be passed. Similarly, in the (-ve) half cycle, D_1 is in reverse bias and D_2 is in forward bias, hence current only from D_2 will be passed. Hence, rectification happens throughout.

(a) Peak Inverse Voltage (PIV) = $2V_m$

(b) $V_{avg} = V_{DC} = \frac{1}{T} \int_0^T V_m \sin(\omega t) dt = \frac{2V_m}{\pi}$

(c) $V_{rms} = \sqrt{\frac{1}{T} \int_0^T V_m^2 \sin^2(\omega t) dt} = \sqrt{\frac{V_m^2}{2}} = \frac{V_m}{\sqrt{2}}$

(d) Ripple factor (RF) = $\sqrt{\left(\frac{V_{rms}}{V_{avg}}\right)^2 - 1} = \sqrt{\left(\frac{V_m/\sqrt{2}}{(2V_m/\pi)}\right)^2 - 1} = \sqrt{\left(\frac{\pi}{2\sqrt{2}}\right)^2 - 1} = 0.482$

(e) Efficiency (η) = $\frac{P_{out}}{P_{in}} = \frac{V_{DC}^2/R}{V_{rms}^2/R} = \left(\frac{V_m}{\frac{2V_m}{\pi}}\right)^2 = \left(\frac{\pi}{2\sqrt{2}}\right)^2 \approx 0.8106 = 81.06\%$

observations and Calculations:

(i) Half Wave Rectifier:

	Theoretical	Practical
Max Voltage (V_m)	4V	3.6V
PIV	4V	3.6V
RMS Voltage (V_{rms})	2V	1.8V
DC Voltage (V_{DC})	1.273V	1.45V
Ripple factor	1.211	1.212
Efficiency (η)	40.53%	40.46%

(ii) Full wave Rectifier:

	Theoretical	Practical
Max Voltage (V_m)	16V	15V
PIV	32V	30V
RMS Voltage (V_{rms})	11.314V	10.606V
DC Voltage (V_{DC})	10.185V	9.549V
Ripple factor	0.483	0.483
Efficiency (η)	81.38%	81.06%

conclusion:

Half wave and Full wave rectifiers have been studied and their characteristics have been observed.

Experiment - 06

24CSE1027

Aim: To study and verify truth tables for various POS and SOP combinations.

Apparatus: AND gate (SN74F08N), OR gate (74HC32N), NOT gate (74SNH04N), digital IC kit, connecting wires.

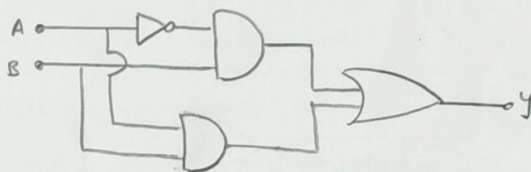
Theory:

→ Logic gate: A logic gate is a device that performs boolean function, a logical operation performed by one or more binary inputs that produces a single binary output.

→ SOP combination: SOP stands for 'Sum of Products'. Here, the inputs are ANDed together and this combination is finally ORed. An example of one such SOP combination is:

$$Y = (\bar{A} \cdot B) + (A \cdot \bar{B})$$

A	B	$\bar{A}$	$\bar{A} \cdot B$	$A \cdot \bar{B}$	Y
0	0	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	1
1	1	0	0	0	0

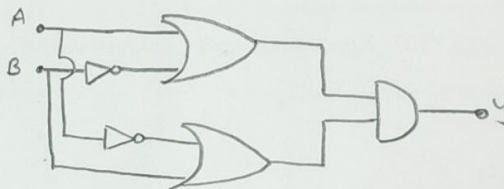


→ POS combination: POS stands for 'Product of Sums'. Here, the inputs are ~~ANDed~~ ORed together and this combination is finally ANDed.

An example of one such POS combination is:

$$Y = (A + \bar{B}) \cdot (\bar{A} + B)$$

A	B	$\bar{A}$	$\bar{B}$	$A + \bar{B}$	$\bar{A} + B$	Y
0	0	1	1	1	1	1
0	1	1	0	1	0	0
1	0	0	1	0	1	0
1	1	0	0	1	1	1



Procedure:

- 1> Place ICs of AND gate, OR gate, and NOT gate on the breadboard of the digital IC kit.
- 2> Make connections of A, B, and Y as per given diagrams.
- 3> Connect ground (GND) to pin 7 and V_{CC} (+5V) to pin 14. It is advised to create a common ground and V_{CC} for all ICs.
- 4> Test the combination and verify if truth table is correct.
- 5> Repeat for other combination.

Conclusion:

SOP and POS combination are studied and verified.

$$(A + B) \cdot (A + \bar{B}) = Y$$

Y	A	B	$A \bar{B}$	$\bar{A} B$	$A + B$	$A + \bar{B}$
0	0	0	0	0	0	0
1	0	1	0	1	1	0
0	1	0	1	0	1	1
1	1	1	1	1	1	1

$$(A + B) \cdot (\bar{A} + \bar{B}) = Y$$

Y	A	B	$A \bar{B}$	$\bar{A} B$	$A + B$	$\bar{A} + \bar{B}$
1	0	0	0	0	0	1
0	0	1	0	1	1	0
0	1	0	1	0	1	0
1	1	1	1	1	1	0